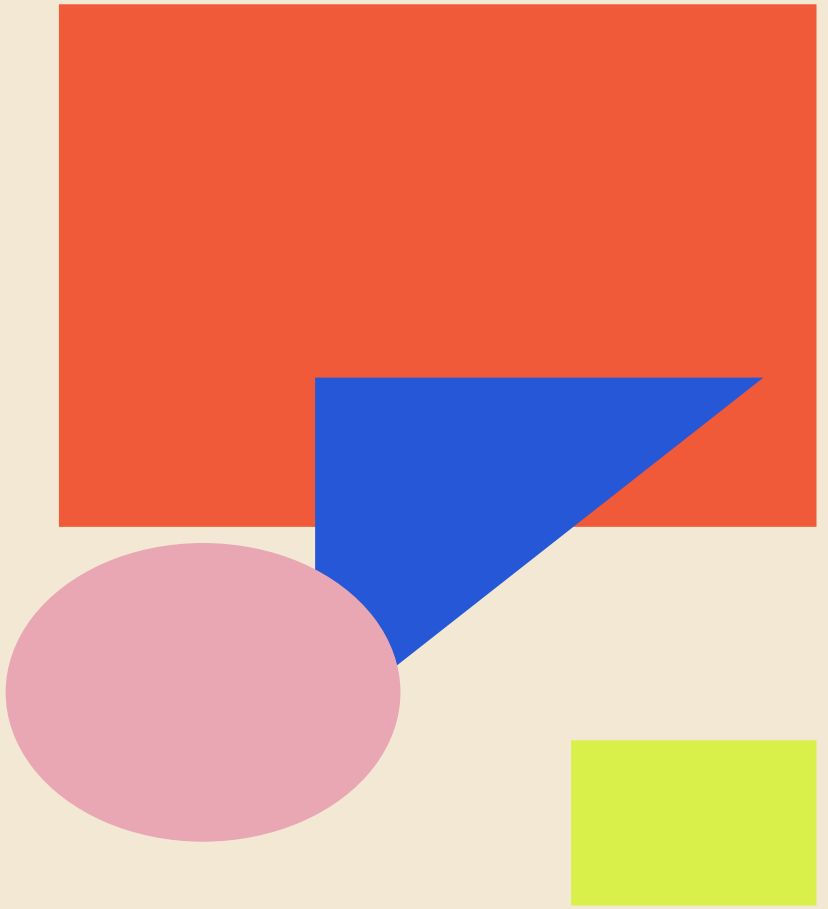
DATA SYSTEMS ADMINISTRATOR · ENTRY-LEVEL COURSE

TROUBLE-SHOOTING 101

Find it. Fix it. Write it down.



COURSE ROUTE

ONE SKILL. SIX STOPS.



hands-on units, one job skill each

AFTER THIS COURSE

You can isolate a fault layer by layer, open any admin tool in seconds, read the logs, and write the ticket.

01 **Think like a troubleshooter**
the six-step method, start to finish

02 **Read the OSI ladder**
layers 1–4, symptom by symptom

03 **Drive the Run menu**
Win+R — the fast lane to every admin tool

04 **Triage with Windows tools**
Task Manager · Services · Event Viewer

05 **Tune Internet Options**
when ping works but the browser doesn't

06 **Log it at the help desk**
record keeping that saves the next tech

UNIT 01 · METHOD

THINK LIKE A TROUBLESHOOTER



A user says: “The internet is broken.”

Where do you even start? This unit gives you a method that always has a next step.

In this unit: the six-step method · three ways to attack a problem · narrowing the blast radius

Six steps. Same order. Every single ticket.

1	**Identify the problem**	Ask: what exactly fails? What changed? Since when?
2	**Establish a theory**	Start with the most likely cause, not the most exotic one.
3	**Test the theory**	Change one thing at a time, so you know what worked.
4	**Plan the fix**	Big or risky changes get a plan before anyone touches anything.
5	**Verify**	Confirm the original problem is gone — and nothing new broke.
6	**Document**	Write down what happened — Unit 06 shows you how.

The industry-standard (CompTIA-style) troubleshooting method — the same loop real help desks run on every ticket.

Top, bottom, or straight through the middle?

Top-down

Start at the app (browser, email, printer software) and work down the ladder toward the cable.

Best when: one app misbehaves and everything else is fine.

Bottom-up

Start at the cable and the link light, then climb the ladder toward the app.

Best when: nothing works at all — a total outage.

Divide-and-conquer

Start in the middle with a ping test, then split the problem in half again and again.

Best when: you're not sure where to start — the pro's default.

Class call-out: I describe a symptom — you call the direction you'd take, and why.

Drill: how big is the blast radius?

Before you touch anything, ask:

- → One machine — or the whole room?
- → One app — or every app?
- → Wired, Wi-Fi, or both?
- → Did it ever work? What changed?

Narrowing the scope is the core diagnostic skill — it turns “everything is broken” into a problem you can actually test.

PRACTICE SCENARIO · ILLUSTRATIVE

Three of thirty lab PCs can't reach the printer. The other 27 print fine. You changed nothing overnight — but the switch was replaced yesterday.

What does the scope tell you — and what is your first test?

Done when: you can say in one sentence what is affected, what is not, and your first test.

UNIT 02 · NETWORK MODEL

READ THE OSI LADDER



Four layers explain almost every network problem you'll meet on the job.

Learn the ladder once, and every symptom gets a home.

In this unit: layers 1–4 · matching symptoms to layers · the four-ping test

Troubleshoot from the bottom rung up.

The OSI model splits networking into 7 layers. On the job, layers 1–4 are where most faults live — so that's where this course stays.



“Is the app's port open, or is a firewall blocking it?”

“Can you ping the gateway? Does a name resolve to an IP?”

“Can devices on the same switch see each other?”

“Is it plugged in? Is the link light on?”

START AT L1 and climb only when the rung below checks out.

Every symptom has a home layer.

No link light, dead port	→	L1	cable, NIC, or the wall jack
Same-switch devices can't see each other	→	L2	switch or VLAN trouble
Can't ping the gateway; names won't resolve	→	L3	IP config, routing, or DNS
One app is blocked, everything else works	→	L4	a firewall rule or a closed port
Ping works, but browsing fails	→	L5+	proxy or browser config — Unit 05

Class drill: your instructor calls a symptom — you call the layer.

Four pings tell you where it broke.

1	ping 127.0.0.1	Your own network stack. Fails → the PC's TCP/IP is broken.
2	ping	Your local network. Fails → cable, Wi-Fi, switch, or IP config.
3	ping 8.8.8.8	The internet by IP. Fails → router, ISP, or gateway config (L3).
4	ping google.com	Adds DNS. Fails while step 3 passes → a DNS problem, not the network.

The first ping that fails — names the layer.

Companion tools: ipconfig (your IP settings, L3) · tracert (the path packets take, L3) · Wireshark (see real packets — advanced demo).

Two myths to drop today.

MYTH 01

“The OSI model is what's actually running on the network.”

Truth: OSI is a reference model — a shared map for talking about networks. The internet actually runs on the simpler TCP/IP model. Use OSI as your troubleshooting map, not as a photograph of reality.

MYTH 02

“Resetting the browser fixes network problems.”

Truth: a browser reset only touches settings above Layer 4. It cannot fix a cable, switch, IP, or port problem. Run the ping ladder first (Unit 02), then tune the browser (Unit 05).

DRIVE THE RUN MENU



Win+R: the fast lane to every admin tool you'll use this semester.

Pros don't hunt through menus — they type three letters and they're in.

In this unit: seven commands worth memorizing · .msc vs .cpl vs .exe

Seven commands worth memorizing.

cmd	**Command Prompt**	ping, ipconfig, tracert — Unit 02's tools
msconfig	**System Configuration**	startup and boot troubleshooting
services.msc	**Services**	start, stop, configure Windows services — Unit 04
eventvwr	**Event Viewer**	the PC's own log book — Unit 04
devmgmt.msc	**Device Manager**	drivers and hardware — Layer 1's home on Windows
ncpa.cpl	**Network Connections**	adapters and IP settings
inetcpl.cpl	**Internet Options**	browser and proxy settings — Unit 05

Your task: start your own cheat sheet this week — add every new run command you meet.

Same tool, three doors.

DEMO — OPEN EVENT VIEWER THREE WAYS

1. **Start menu → type “Event Viewer”**
2. **Win+R → eventvwr**
3. **Command line → eventvwr**

All three open the exact same console. The Run box is simply the fastest door — and it works even when the Start menu is the thing that's broken.

DECODER — WHAT THE ENDINGS MEAN

.msc	management consoles (services.msc)
.cpl	Control Panel applets (ncpa.cpl)
.exe	a program — notepad, cmd

MYTH BUSTED

The Run box doesn't just launch programs — it opens management consoles and control panels too.

TRIAGE WITH WINDOWS TOOLS



Task Manager, Services, and Event Viewer.

See what's running, what's stopped, and what went wrong — in that order.

In this unit: find the resource hog · restart a service · read the logs

Task Manager: who's eating the PC?

Open it: **Ctrl+Shift+Esc** — or type **taskmgr** in the Run box (Unit 03).

1	**Processes tab**	Sort by CPU, Memory, or Disk to find the resource hog.
2	**Startup tab**	Disable high-impact apps that slow every boot.
3	**End task**	The safe way to kill a frozen app — after you've looked.

MYTH CHECK — BASELINE FIRST

High CPU isn't always bad. A busy PC isn't automatically a broken PC — know the machine's normal baseline first, then compare.

Services: the apps with no window.

Open it: Win+R → **services.msc** · right-click a service to start, stop, or restart it.

STARTUP TYPES — WHEN DOES IT RUN?

Automatic — starts with Windows, always on

Automatic (Delayed) — starts shortly after boot

Manual — runs only when something asks for it

Disabled — never runs (don't set this if you're unsure)

Before you stop anything: services depend on each other. Check the Dependencies tab first — stopping one service can quietly break three others.

CLASS DEMO

Stop the **Print Spooler** service.

Try to print — it fails.

Restart the service — printing returns.

Nothing changed except the service. That's how you prove a cause instead of guessing one.

Event Viewer: the PC's flight recorder.

Open it: Win+R → **eventvwr** → Windows Logs

THE THREE LOGS THAT MATTER

Application	app crashes and errors
System	Windows itself: drivers, services, boot
Security	logons and access attempts

HOW TO READ AN EVENT

Level — Information, Warning, Error, Critical

Event ID — the error's fingerprint; search it

Time — line events up with the incident

Filter — hide the noise, keep Errors only

Logs are evidence — not noise.

To reconstruct an incident: open the System log, filter to the time things went wrong, and read upward from the first Error. You'll do exactly this in the capstone — and quote it in your log book (Unit 06).

UNIT 05 · BROWSER SETTINGS

TUNE INTERNET OPTIONS



inetcpl.cpl — where browser, cache, and proxy settings live.

The network can be perfect while the browser still fails. This is where you look next.

In this unit: the six tabs · the bogus-proxy pitfall · when a reset helps (and when it can't)

Ping works, but the browser doesn't? Look here.

Open it: Win+R → **inetcpl.cpl** — the six tabs below.

General	homepage and cached files (clear cache)
Security	trust zones for websites
Privacy	cookies and pop-up blocking
Connections	LAN settings — **the proxy lives here**
Programs	default browser and add-ons
Advanced	deep settings and the big Reset button

THE CLASSIC TICKET — CLASS DEMO

A bogus proxy entry breaks all browsing — sometimes left behind by malware.

Fix: Connections → LAN settings → uncheck the proxy server.

Browser problems live above Layer 4.

If the ping ladder (Unit 02) passes, suspect DNS → proxy → browser config, in that order.

Myth check: “Reset Internet settings” can't fix a cable, switch, or IP problem — run the ladder first.

UNIT 06 · HELP DESK

LOG IT AT THE HELP DESK



The fix isn't finished until it's written down.

Remember step 6 of the method? This unit is where “document” becomes a habit.

In this unit: the six-line log entry · log book vs. ticket system · clean handoffs to tier 2

A good log entry has six lines.

1 · WHO — who reported it, and where

2 · WHEN — date and time it happened

3 · SYMPTOM — in the user's own words

4 · TRIED — every test and change you made

5 · RESULT — what each attempt did

6 · NEXT STEP — fixed, monitoring, or escalate

Escalation etiquette: before handing a ticket to tier 2, record everything you tried — so nobody repeats your work.

EXAMPLE ENTRY · ILLUSTRATIVE

MAR 4 · 10:15 — J. Rivera, Room 204

“Wi-Fi drops every few minutes.”

Pinged gateway: OK. Pinged 8.8.8.8: drops.

Replaced patch cable at wall jack → stable 2 hrs.

Next: monitor tomorrow; escalate if it recurs.

A ticket system is a log book with search.

Same six lines, plus fields: category, priority, status, resolution.

The knowledge loop: today's log entry is tomorrow's first-line fix.

TRANSFER · YOUR FIELD KIT

CARRY THESE FOUR THINGS.

THE 6 STEPS	identify → theory → test → plan → verify → document
THE PING LADDER	127.0.0.1 → gateway → 8.8.8.8 → google.com
THE RUN COMMANDS	cmd · services.msc · eventvwr · inetcpl.cpl (+3 more in Unit 03)
THE 6-LINE LOG	who · when · symptom · tried · result · next step

That's the whole course on one card: a method, a ladder, a toolbox, and a habit. Everything else is practice.

KEEP A LOG BOOK AT EVERY LAB — START THIS WEEK